Algebra and Trigonometry
MATH 1450-02

Name (Print): ____________________

- This exam contains 15 pages (including the front and back of this cover page) and 17 problems. Check to see if any pages are missing.
- This is the first exam and will count for 20% of your final grade. You have 120 minutes (5:00 pm - 7:00 pm) to complete this exam. You may use one index card of notes and a calculator on this exam, but no other outside sources may be consulted.

The following rules apply:

- Every problem on this exam, other than true-or-false problems, requires work of some form. An incorrect answer supported by substantially correct calculations and explanations will receive partial credit. **A correct answer that has no supporting work will not receive full credit.**
- Organize your work, in a reasonably neat and coherent way, in the space provided. Work scattered all over the page without a clear ordering will receive very little to no credit.
- There are 100 points total in this exam, as well as 3 bonus questions worth a total of 6 possible points.

Page:	3	4	5	6	7	8	9	10	11	12	13	Total
Points:	20	10	5	10	10	10	5	5	10	10	5	100
Score:												

Bonus:	

1. Mark each statement as $\boxed{\text{True}}$ or $\boxed{\text{False}}$ by completely shading in the box corresponding to your answer.

2 pts (a) If $f(x) = (2x+2)^2$, then $f(x+h) = (2x+2)^2 + h$.

True False

2 pts (b) The slope of the line that passes through $(6, -6)$ and $(2, -4)$ is -2

True False

2 pts (c) The x-intercept of $y = -4x + 6$ is $\left(\frac{3}{2}, 0\right)$ and the y-intercept is $(0, 6)$.

True False

2 pts (d) The average rate of change of $g(x) = 5x^2 - 5$ on the interval $[x, x+h]$ is $5h + 10x$.

True False

2 pts (e) If $f(x) = \sqrt{x} + 5$ and $g(x) = x^2 + 2$, then the composition $(g \circ f)(x) = g(f(x))$ is given by $(g \circ f)(x) = \sqrt{x^2 + 2} + 5$.

True False

2 pts (f) In interval notation, the domain of the function $f(x) = \frac{4x+4}{\sqrt{6-x}}$ is $(-\infty, 6)$.

True False

2 pts (g) The function $f(x) = \frac{1}{x} + 2x$ is an even function.

True False

2 pts (h) If $f(x)$ is a one-to-one function and $f^{-1}(-5) = -4$, then $f(-4) = -5$.

True False

2 pts (i) The graph of $y = f(x) + 4$ is a horizontal translation of the graph of $y = f(x)$

True False

2 pts (j) If the graph of $y = f(x)$ is the same as the graph of $y = f(-x)$, then $f(x)$ is an even function.

True False

2. Suppose the functions p and m are given by

$$p(x) = \frac{1}{\sqrt{x}}, \text{ and } m(x) = x^2 - 4.$$

Find the domain of the functions given below. Express your answer in interval notation

$\boxed{5 \text{ pts}}$ (a) $\dfrac{p(x)}{m(x)}$

(A) $(0, 2) \cup (2, \infty)$ (B) $(-\infty, -2] \cup [2, \infty)$ (C) $(-\infty, 2) \cup (2, \infty)$ (D) $(-\infty, \infty)$

$\boxed{5 \text{ pts}}$ (b) $p\big(m(x)\big)$

(A) $(-\infty, \infty)$ (B) $(-2, 2)$ **(C)** $(-\infty, -2) \cup (2, \infty)$ (D) $(2, \infty)$

5 pts

3. Solve the inequality

$$3|v+2| - 4 \geq 8.$$

Express your answer in interval notation.

(A) $(-\infty, -6) \cup (2, \infty)$ (B) $(-\infty, \infty)$ (C) $[-6, 2]$ **(D)** $(-\infty, -6] \cup [2, \infty)$

5 pts

4. Find the value of k if the graph of a linear function $y = f(x)$ has slope $m = 3$ and passes through the points $(4, 7)$ and $(k, 1)$.

(A) $k = 2$ (B) $k = -2$ (C) $k = -6$ (D) $k = 6$

5 pts

5. Write the equation of the line that passes through the points given below. Express your answer in slope-intercept form (i.e. in the form $y = mx + b$).

$$(r, k) \text{ and } (r + 2, k + 4)$$

(A) $y = -2x + (2r - k)$ **(B)** $y = 2x + (k - 2r)$ (C) $y = 2x + (2k - 2r)$ (D) $y = 2kx + kr$

5 pts

6. Write an equation for the line perpendicular to $y = 2x - 1$ which passes through the point $(4, 1)$.

(A) $y = \frac{1}{2}x + 3$ (B) $y = -2x + \frac{1}{3}$ (C) $y = -\frac{1}{2}x + \frac{3}{2}$ **(D)** $y = -\frac{1}{2}x + 3$

5 pts

7. Suppose a town has a population of $1,000$ in the year 1999, and suppose the population of the town grows at a constant rate of 125 per year. Find the equation of the line that models the town's population P as a function of t, where t is the number of years since the model began.

(A) $P = 125(t - 1999) + 1000$ (B) $P = -125t + 1000$ (C) $P = 125t + 1250$. **(D)** $P = 125t + 1000$

8. Perform the following operations and express the results as a simplified complex number (i.e. your answer should be a complex number of the form $a + bi$ where a and b are real numbers):

5 pts (a) $(4 + i)(2i)$

(A) $2 - 8i$ (B) $8 - 2i$ **(C)** $-2 + 8i$ (D) $2 + 8i$

5 pts (b) $\dfrac{4 - 2i}{2 + 2i}$

(A) $\dfrac{1}{2} - \dfrac{3}{2}i$ (B) $2 - \dfrac{3}{2}i$ (C) $\dfrac{1}{2} + \dfrac{2}{3}i$ (D) $\dfrac{3}{2} - \dfrac{1}{2}i$

5 pts 9. Solve the following equation over the complex numbers:

$$x^2 - 2x + 10 = 0.$$

(A) $x = 3 \pm i$ (B) $x = 1 \pm 3i$ (C) $x = \frac{1}{3} \pm 3i$ (D) $x = 1 \pm \frac{1}{3}i$

5 pts 10. Use the information given below about the graph of a polynomial function $f(x)$ to determine the function. Assume the leading coefficient is 1 or -1.
HINT: Use the vertex form of a quadratic function $y = a(x-h)^2 + k$

(1) The y-intercept is $(0, 49)$
(2) The x-intercepts are $(-7, 0)$ and $(7, 0)$
(3) As $x \to -\infty$, $f(x) \to -\infty$
(4) As $x \to \infty$, $f(x) \to -\infty$
(5) The degree of $f(x)$ is 2

A $f(x) = x^2 - 49$ **B** $f(x) = -x^2 + 49$ **C** $f(x) = -(x-7)(x+7)$ D $f(x) = (x-7)(x+7)$

$\boxed{5 \text{ pts}}$ 11. Find the zeros of the following function $f(x)$ and give the multiplicity of each:

$$f(x) = (x^3 - 5x^2 + 6x)(12x^3 - 4x^4).$$

$\boxed{5 \text{ pts}}$ 12. Determine the end behavior of the function $f(x)$ given by

$$f(x) = -4x^4 - 3x^2 + 3x - 2.$$

13. Suppose $f(x)$ and $g(x)$ are the polynomial functions given by

$$f(x) = x^3 + 4x^2 + 5x + 2, \text{ and } d(x) = x + 2.$$

5 pts

(a) Find polynomials $q(x)$ and $r(x)$ such that $f(x) = d(x)q(x) + r(x)$, where $r(x)$ is either 0 or a polynomial of degree strictly less than the degree of $d(x)$.

(A) $q(x) = x^2 + 2x + 1$, $r(x) = 0$

(B) $q(x) = x^3 + 3x^2 + 5x + 6$, $r(x) = x - 2$

(C) $q(x) = x^3 + 3x^2 + 4x + 4$, $r(x) = x - 1$

(D) $q(x) = x^3 + 3x^2 + x - 2$, $r(x) = x + 2$

5 pts

(b) Using the Remainder Theorem and your answer from part (a), compute $f(-2)$ **without** explicitly evaluating $f(-2)$ (that is, *without* plugging in -2 for x in $x^3 + 4x^2 + 5x + 2$ and solving for a value).

(A) $f(-2) = -2$

(B) $f(-2) = 2$

(C) $f(-2) = 0$

(D) $f(-2) = 1$

5 pts 14. Suppose $f(x)$ is the polynomial function

$$f(x) = x^3 - x^2 - 26x + 20.$$

Given that $(x+5)$ is a factor of $f(x)$, use the factor theorem to determine all of the real zeros of $f(x)$.

(A) $x = -5$, $x = 3 + \sqrt{5}$, $x = 3 - \sqrt{5}$

(B) $x = -5$, $x = 3 + \sqrt{5}$, $x = -3 + \sqrt{5}$

(C) $x = 5$, $x = \sqrt{5} + 3$, $x = \sqrt{5} - 3$

(D) $x = \sqrt{5}$, $x = 3$, $x = -3$

$\boxed{\text{2 (bonus)}}$ 15. Suppose the rational function $f(x)$ is given by $f(x) = \dfrac{5x+7}{2x+4}$. Find $f^{-1}(x)$.

$\boxed{\text{2 (bonus)}}$ 16. Suppose $f(x)$ is the rational function given by

$$f(x) = \frac{x^2+3x+2}{x^2-5x+6}.$$

Find the vertical asymptotes of $f(x)$.

2 (bonus) 17. Find the inverse of the function $f(x) = x^2 + 4x + 3$ over the domain $[-2, \infty)$. The domain of the inverse function will be $[-1, \infty)$.